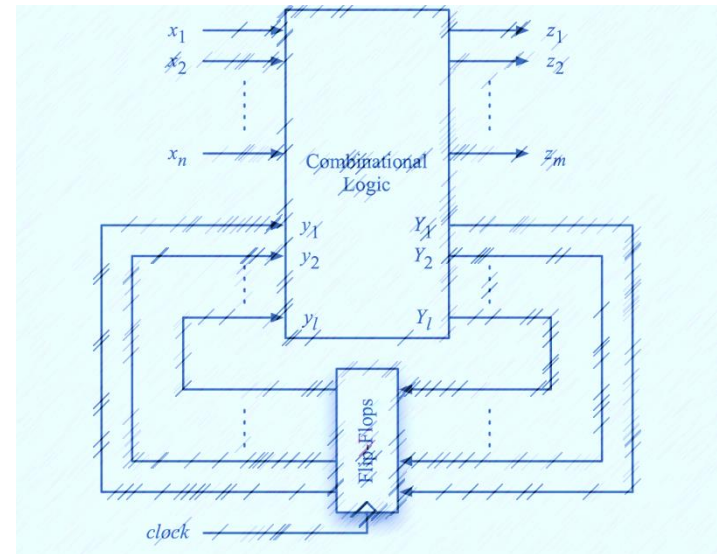


Sequential ATPG

- Introduction
- Time-frame expansion methods
- Simulation-based methods*
- Issues of Sequential ATPG* (not in exam)
- Conclusions



Issues of Sequential ATPG

- ① Ckts. **without initialization** input
- ② **Potentially detected faults** can escape test
- ③ **Asynchronous** ckt. requires special attention

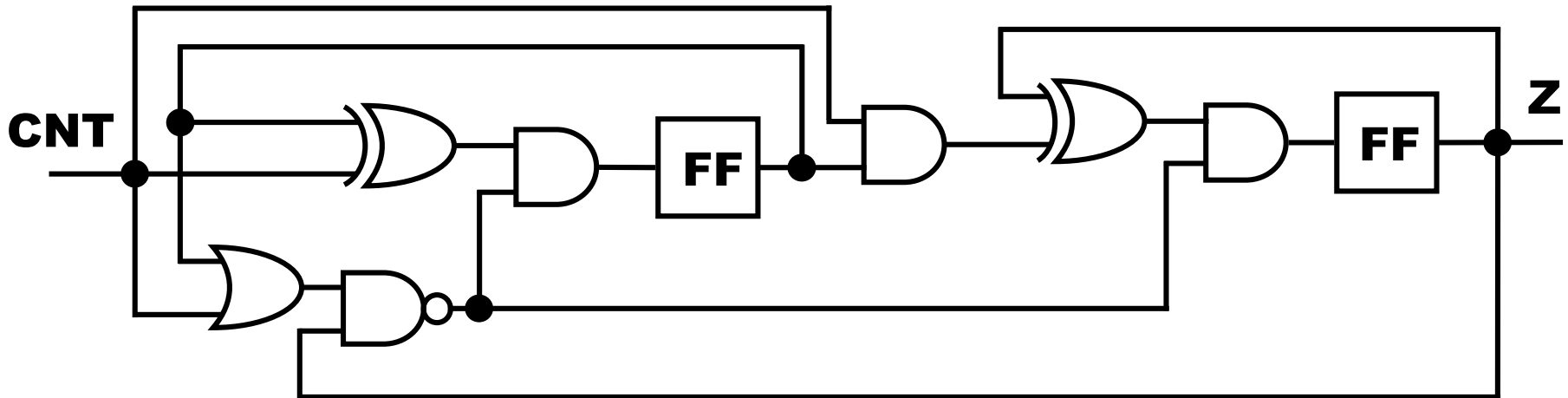
①

Circuit w/o Initialization Input

- Example

- ◆ Mod 3 counter (BA Fig. 8.12)

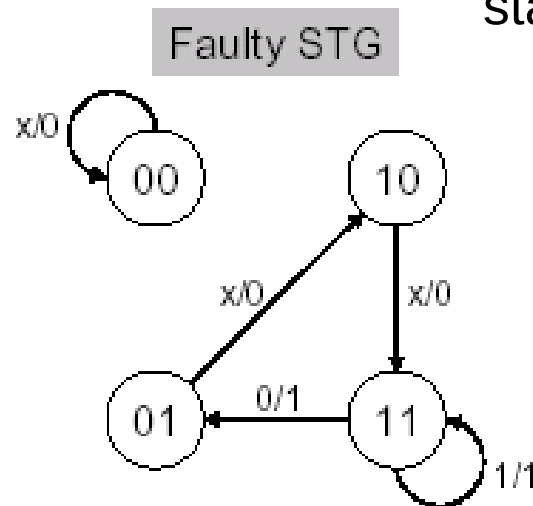
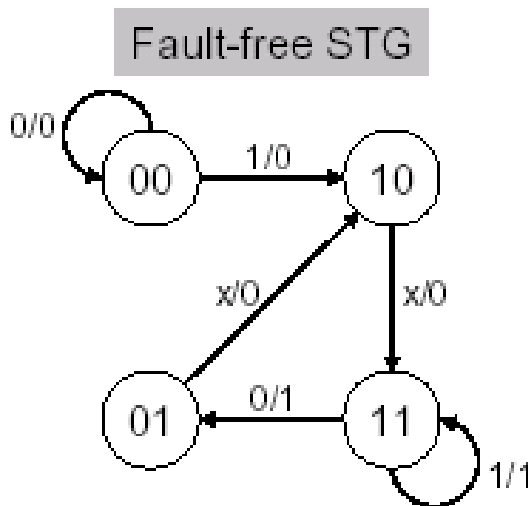
- 10 → 00 → 01 → 10 →



Cannot Find Initialization Sequence

② Potentially Detected Faults

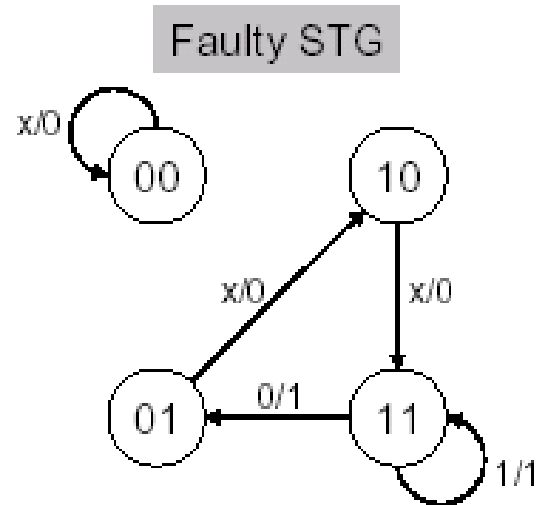
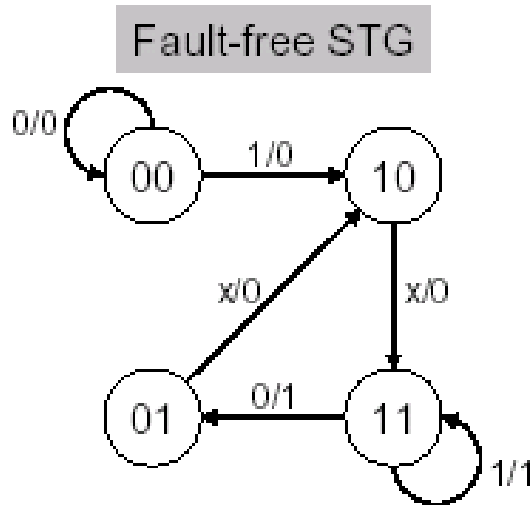
- **DEF:** faults that may or may not be detected
 - ◆ Also see Video 5.7
- **Example:** detection of fault depends on the power-up states
 - ◆ $Y_1 = xy_1y_2 + y_1y_2' + y_1'y_2$ (+ $xy_1'y_2'$) faulty STG missing term
 - ◆ $Y_2 = y_1y_2 + y_1y_2'$
 - ◆ $Z = y_1y_2$



state transition graph

X/Z
(Input/output)
 y_1y_2
(states)

X/Z



- {0 1 1} initializes the circuit to state 11
- Good response to input {0110} is
 - ◆ z = 1001

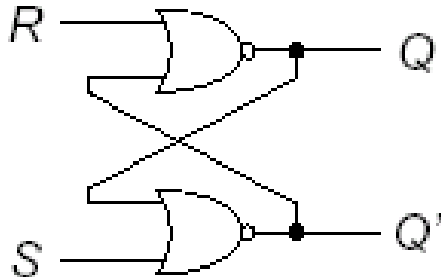
- {0 1 1} cannot initialize circuit
- Response at z to {0110}:
 - ◆ Power-up state 00,
 - z = 0000
 - fault detected
 - ◆ Power-up state 01,10, or 11
 - z = 1001
 - fault **NOT** detected

Fault Detection is Uncertain

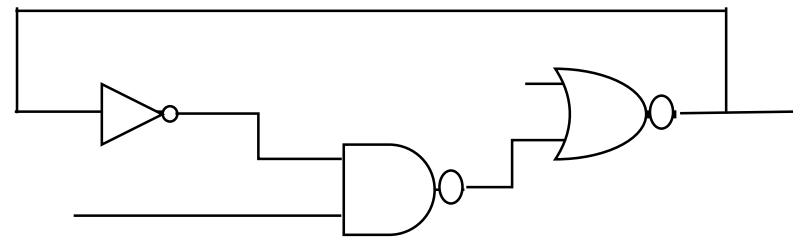
③ Asynchronous Circuits

- No explicit clock. Signals can change asynchronously
- **Timing** can be difficult to model
 - ◆ Test patterns generated may cause *rices* and *hazards*
 - ◆ Need to verify test sequence with a fault simulator
- Example: Circuits with combinational loop

local feedback



global feedback



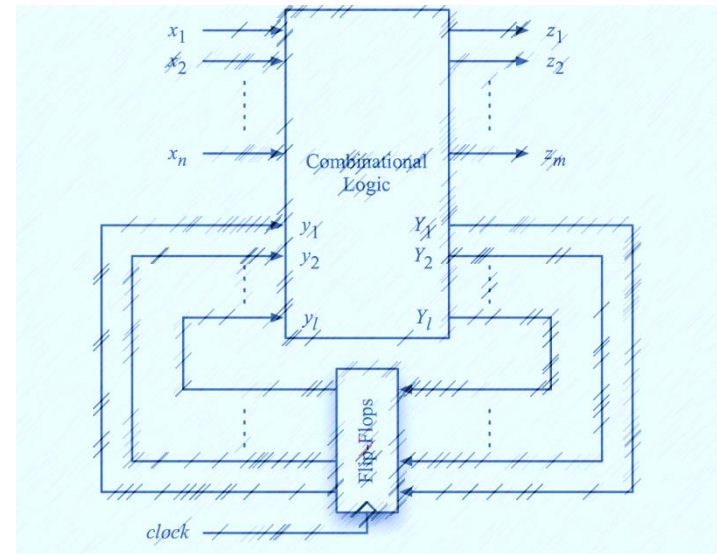
QUIZ

Q: Suppose you are a designer. Which of the following circuits you should avoid in order to avoid sequential ATPG?

- A) Combinational circuits with feedback loops**
- B) Non-scan flip-flops without reset pins (video 11.8)**
- C) Circuits with many different types of flip-flops**
- D) Circuits with SRAM memories**

Sequential ATPG

- Introduction
- Time-frame expansion methods
- Simulation-based methods*
- **Issues of Sequential ATPG***
- **Conclusions**



Concluding Remarks

- Sequential ATPG
 - ◆ Generate PI patterns, observe PO. **No scan allowed**
- Benefits
 - ◆ Enable **at-speed testing**
 - ◆ Handles **partial scan or non-scan** circuits
- Problems
 - ◆ **Low fault coverage, long run time, large memory**
- Techniques
 - ◆ **Time frame expansion** (EBT, BACK, Extended-D ...)
 - ◆ **Simulation** (CONTEST)
- Current DFT & ATPG Practice
 - ◆ Use as much **combinational ATPG** as possible
 - ◆ Use sequential ATPG **only when necessary**

Sequential ATPG is Difficult. Need DfT! (Ch11)

References

- [Agrawal 88] Agrawal, Vishwani D., K-T. Cheng, and Prathima Agrawal. "CONTEST: A concurrent test generator for sequential circuits," Design Automation Conference, 1988. Proceedings., 25th ACM/IEEE. IEEE, 1988.
- [Cheng 88] Cheng, W-T. "The BACK algorithm for sequential test generation," Computer Design: VLSI in Computers and Processors, 1988. ICCD'88., Proceedings of the 1988 IEEE International Conference on. IEEE, 1988.
- [Holland 1975] Holland, J.H., 1975. Adaptation in Natural and Artificial Systems. MIT Press, Cambridge, MA.
- [Kubo 68] H. KUBO "A procedure for generating test sequences to detect sequential circuit failures," NEC Res. & Dev., (Oct. 1968), 69 –78.
- [Marlett 78] Marlett, Ralph A. "EBT: A comprehensive test generation technique for highly sequential circuits," Proceedings of the 15th Design Automation Conference. IEEE Press, 1978.
- [Muth 76] P. Muth "A nine-valued circuit model for test generation," IEEE Trans. Comput., 25 (6), pp. 630–636, 1976.
- [Huffman 53] D. A. Huffman, "The Synthesis of Sequential Switching Circuits," MIT Thesis, 1953.

Commercial Tools

- **Mentor Graphics**
 - ◆ **Flextest**
- **Synoposys**
 - ◆ **Tetramax**
- **Syntest**
 - ◆ **turboscan**

Academic Tools

- Time-Frame Expansion
 - ◆ ESSENTIAL [89]
 - ◆ FASTEST[89]
 - ◆ HITEC [Niermanh & Patel 91]
 - ◆ Lee-Reddy [91]
- Genetic Algorithm
 - ◆ CRIS [Saab & Abraham 96]
 - ◆ GATEST [Rudnick 97]
 - ◆ GATTO [96]
 - ◆ STRATEGATE [97]